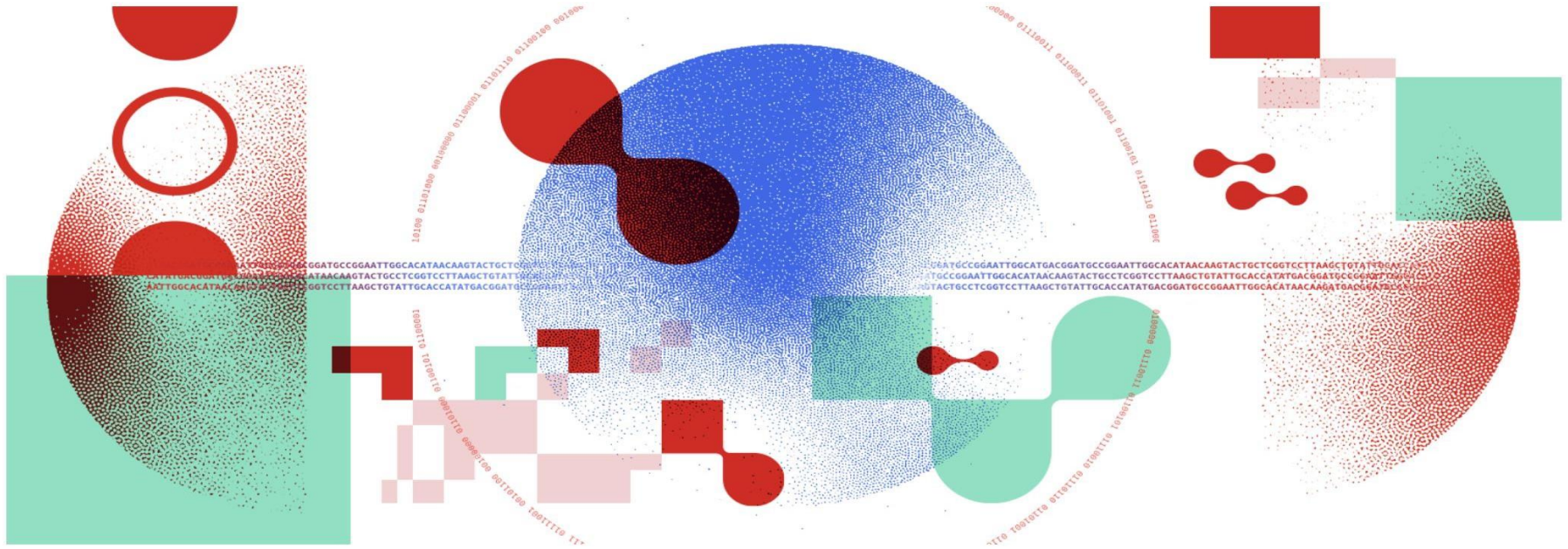




Advanced Statistical Modelling

Lausanne, September 2025

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Logistic Regression and GLM



Statistical Models

Are used for explanation and prediction

Statistical models predicts the mean Y for any combination of predictors.

General form: $g(Y) = f(X) (+ \text{Error})$

with a stochastic process ($\Leftrightarrow$ errors)

Y : dependent variable
(response variable,
observed outcome)

X : independent/
explanatory variable(s)
(grouping variable,
predictor)

Types of response and predictors variables

- binary (2 groups)

(e.g. yes/no, passed/failed, male/female, ill/healthy, responder/NR, ...)

- categorical (k groups)

(e.g. phenotype, genotype, degree of physical activity, ...)

- continuous (i.e. pot. infinite number of groups)

(e.g. weight, blood pressure, gene expression value, ...)

- Censored data (e.g. survival data for patients, living beings, technical devices, ...)

Types of variables

Response variable's type determines the suitable regression method(s) :

continuous response -> Linear regression

binary response -> **Logistic regression**

count response -> Poisson regression

Logistic regression

What is Logistic Regression?

Form of regression that allows the prediction of discrete variables by a mix of continuous and discrete predictors.

Discrete $\sim$ continuous / discrete

Example: Responder status $\sim$ dosis

Binary Logistic Regression Model

Y = Binary **response**, ex. **Responder status (1 / 0)**

X = Quantitative **predictor**, ex. Dose, genotype for gene G, ...

π = Proportion / **Probability** of »event 1« at any X

Given π we assume that a (always identical) stochastic process “determines” the event(s) observed outcome.

For a group with the same configurations of X , the same π :
there is a **binomial** distribution $B(n,p)$ of »event 1«
 n = number of observations at this X ,
 p = prob. of event 1 and

Proportion of “success”

In linear regression the model predicts the mean Y for any combination of prediction (the $E[Y | X]$). What's the mean of a 0/1 indicator variable?

*The **Proportion** of “cases 1” among n observations.*

$$\pi = \bar{y} = \frac{\sum y_i}{n}$$

***Goal of logistic regression:** Predict the “**true**” probability of success, π , at any value of the predictor(s).*

Modeling overview revised

$X \rightarrow E(Y)$ the expected value of $Y \rightarrow$ individual values Y_i

stochastic error
(lm: normal; log: binomial)

now a prob. $[0,1]$

Deterministic function

(lm: Linear / affine function or polynomial etc. , linear in parameters

glm a new approach, linear in parameters, but with a transformation linking it to the $E(Y)$)

Relation probability – odds

$$\mathbf{odds} = \frac{\pi}{1 - \pi} \Leftrightarrow \pi = \frac{\mathbf{odds}}{1 + \mathbf{odds}}$$

π in $[0, 1]$, odds in $(0, +\infty)$,

$\pi = 0.5$ odds = 1

$\pi = 0.9$ odds = 9

$\pi = 0.1$ odds = $1/9 = 0.111$

not symmetric

Logistic curve

Logit is the **logarithm of the odds**
(**log=ln=log_e**)

$$\log\left(\frac{\pi}{1-\pi}\right)$$

Probability of success

Probability of failure

$$\pi = 0.50 \Leftrightarrow \text{logit} = 0$$

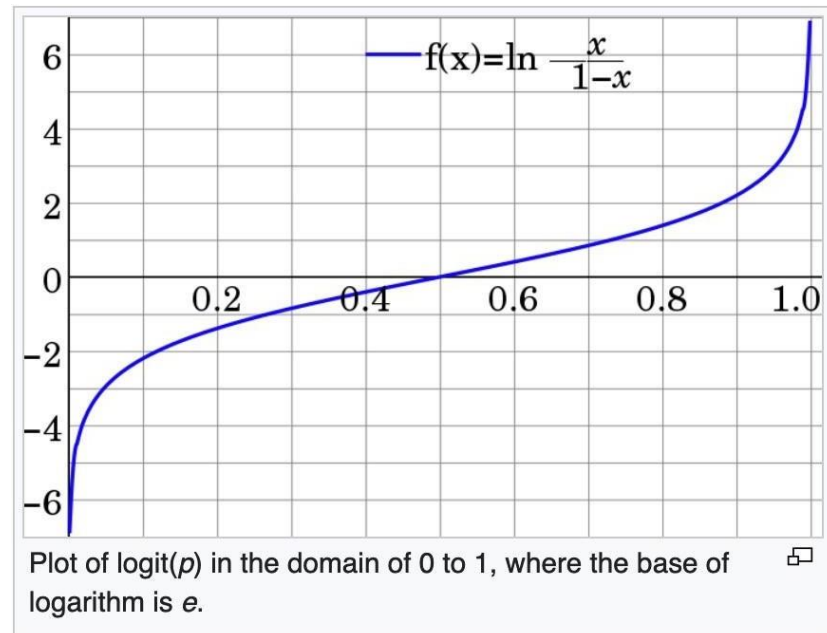
$$\pi = 0.70 \Leftrightarrow \text{logit} = 0.84$$

$$\pi = 0.30 \Leftrightarrow \text{logit} = -0.84$$

$$\pi \rightarrow 1 \Leftrightarrow \text{logit} \rightarrow \text{inf}$$

$$\pi \rightarrow 0 \Leftrightarrow \text{logit} \rightarrow -\text{inf}$$

Symmetric, range (-inf, +inf)



Binary Logistic Regression Model

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X$$

The predictors acts at the level of the log odds

$X \rightarrow \text{logit} \rightarrow \underset{\text{prob. [0,1]}}{E(Y)} \rightarrow \text{individual values } Y_i$

**Deterministic function,
Linear in the parameters describing the effects of the expl. Vars.**

The logit is called a **link function**, links the level of the observed events (**response level**) to the level at which the predictors effects are acting (**link level**)

GLM LOGISTIC

Predictors $X \Rightarrow E(\text{logit}(\pi) \mid X) \Rightarrow$ observations Y_i

$\Rightarrow$ The assumed
model of effects

$\Rightarrow$ The assumed underlying
stochastic process
(generating the data)

$\Rightarrow$ ex. Logit
is linear in β 's

$\Rightarrow$ ex. Binomial distribution

Binary Logistic Regression Model

$$\textit{odds} = \frac{\pi}{1 - \pi} \Leftrightarrow \pi = \frac{\textit{odds}}{1 + \textit{odds}} = \frac{1}{1 + 1/\textit{odds}}$$

Logit form

$$\log\left(\frac{\pi}{1 - \pi}\right) = \beta_0 + \beta_1 X$$

Link - Level

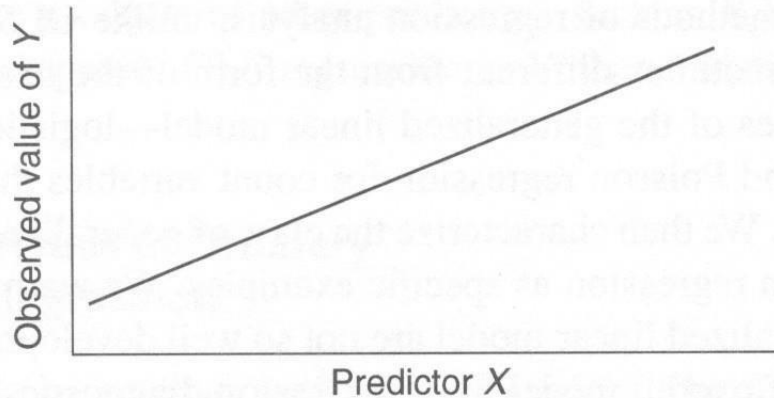
Probability form

$$\pi = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

Response - Level

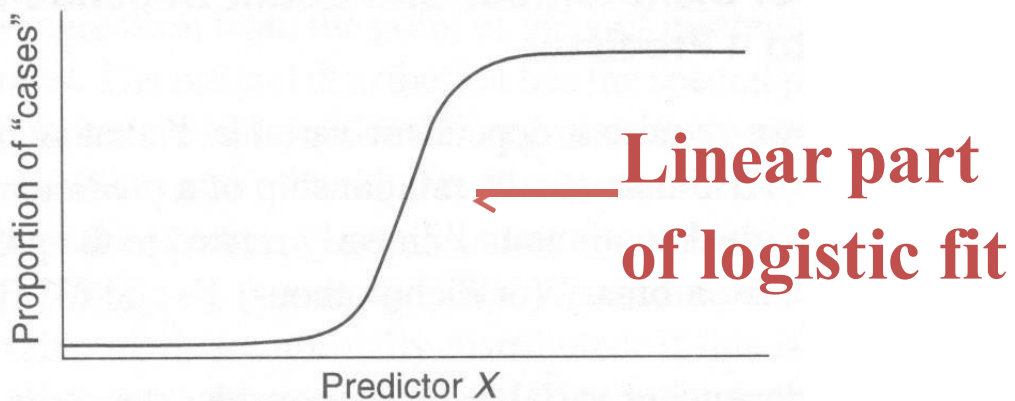
The logistic function

(A) For a continuous outcome variable Y , the numerical value of Y at each value of X .



Change in probability is not constant (linear) with constant changes in X

(B) For a binary outcome variable, the proportion of individuals who are “cases” (exhibit a particular outcome property) at each value of X .



$$\pi = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Odds for X: $odds = e^{\beta_0 + \beta_1 X}$

Odds for X+1: $odds = e^{\beta_0 + \beta_1 (X+1)}$

Odds ratio (odds for X+1 / odds for X):

$$\frac{e^{\beta_0 + \beta_1 (X+1)}}{e^{\beta_0 + \beta_1 X}} = e^{\beta_0 + \beta_1 (X+1) - (\beta_0 + \beta_1 X)} = e^{\beta_1}$$

We increase X_1 by one unit (**+1, additive**)

The log odds is increased by β_1 (additive)

The odds is increased by a factor $\exp(\beta_1)$ (**multiplicative**)

The probability is increased by ? (**question!**)

Assumptions

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 X$$

The logistic model assumes a linear relationship between the *predictors* and the *log(odds)*.

$$odds = \frac{\pi}{1-\pi} = e^{\beta_0 + \beta_1 X}$$

Logistic regression is a special case of a Generalized Linear Model GLM

Ordinary Least Squares regression provides linear models of continuous variables. However, much data of interest to statisticians and researchers are not continuous and so other methods must be used to create useful predictive models.

GLM LOGISTIC

Predictors $X \Rightarrow E(\text{logit}(\pi) \mid X) \Rightarrow$ observations Y_i

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model of effects

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stochastic process
(generating the data)

$\Rightarrow$ ex. Logit
is linear in β 's

$\Rightarrow$ ex. Binomial distribution

GLM Poisson

Predictors $X \Rightarrow E(\text{link}(Y) | X) \Rightarrow$ observations Y_i

$\Rightarrow$ **The assumed
model of effects**

$\Rightarrow$ **stochastic process**

Log (λ_i) linear in β 's

Data $Y_i \sim$ Poisson distribution

Poi (mean $\lambda = E[Y]$)

Stdev = sqrt (λ)

The dispersion is the one expected for a »pure random sampling« that is without any factor of variability increasing the dispersion.

Stddev = sqrt(mean)

ML-estimation, deviance, LRT,
Wald test on coefficients etc:
Like Logistic Regression

Poisson regression

- Basic standard model used for Count data
- Distribution: Poisson, (Restriction: mean = variance : $E(Y)=V(Y)=\lambda$)
- Default Link Function: log link:

$$\ln(\lambda) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$$

$$\Rightarrow \lambda(X_1, \dots, X_k) = e^{\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k}$$

Tests are conducted as in Logistic regression

Poisson regression - assumptions

- Poisson at the Response Level : the response variable is a count per unit of time or space, described by a Poisson distribution.
- Linearity at Link Level : the log of the mean rate, $\log(\lambda)$, is be a linear function of the predictor x .
- Independence: the observations are independent of one another.
- Mean=Variance: the mean of a Poisson random variable is equal to its variance.

Generalized Linear Models

Generalized linear models are fit using the `glm()` function. The form of the `glm` function is

`glm(formula, family=familytype(link=linkfunction), data=)`

Family	Default Link Function
binomial	(link = "logit")
gaussian	(link = "identity")
Gamma	(link = "inverse")
inverse.gaussian	(link = "1/mu^2")
poisson	(link = "log")
quasi	(link = "identity", variance = "constant")
quasibinomial	(link = "logit")
quasipoisson	(link = "log")

How to find the "best fit"

Standard method: **maximum likelihood estimation MLE**

Probability of observations

(Likelihood of the model given the data) = maximum

The **MLE** is the preferred method for statisticians in many situations. It has a series of good properties (best method given some criteria).

Maximum likelihood

- Likelihood measures how well a statistical model explains the observed data.
- For data $y_1, y_2, \dots, y_n$ and parameters θ :
$$L(\theta) = P(Y_1=y_1, \dots, Y_n=y_n \mid \theta)$$

where P is the **probability distribution of your model** (PDF/PMF)

- Probability: parameters fixed $\rightarrow$ ask about data
- Likelihood: data fixed $\rightarrow$ ask about parameters

Example: 3 consecutive coin tosses (H, T, H) with $p = P(\text{Heads})$:

$$L(p) = p^2 (1 - p)$$

Question (challenge): what would be the likelihood of a linear model $Y = \beta_0 + \beta_1 X$

Maximum likelihood

Model	Data Type	Distribution (PDF/PMF)	Form of Likelihood 
Linear regression	Continuous Y	Normal: $Y_i \sim N(X_i\beta, \sigma^2)$	Product of normal densities
Logistic regression	Binary $Y \in \{0, 1\}$	Bernoulli: $P(Y_i = 1) = \pi_i$ with $\pi_i = \frac{1}{1+e^{-X_i\beta}}$	Product of Bernoulli pmfs
Poisson regression	Count $Y \in \{0, 1, 2, \dots\}$	Poisson: $Y_i \sim \text{Poisson}(\mu_i)$, $\mu_i = e^{X_i\beta}$	Product of Poisson pmfs
Binomial regression	Proportion / # successes out of n	Binomial: $Y_i \sim \text{Binomial}(n_i, \pi_i)$	Product of binomial pmfs
Survival (Exponential)	Time-to-event	Exponential: $f(t) = \lambda e^{-\lambda t}$	Product of exponential pdfs
Survival (Weibull)	Time-to-event	Weibull: $f(t) = \frac{k}{\lambda} \left(\frac{t}{\lambda}\right)^{k-1} e^{-(t/\lambda)^k}$	Product of Weibull pdfs

Maximum likelihood

- Goal: Find parameter values θ that maximize the likelihood of the data
 $\hat{\theta} = \operatorname{argmax}_{\theta} L(\theta)$
- Often use the log-likelihood for easier math:
 $\ell(\theta) = \log L(\theta)$

Intuition:

- Choose parameters that make the observed data most plausible.
- Coin toss $\rightarrow$ estimate probability of heads
- Regression $\rightarrow$ estimate coefficients

MLE gives consistent, efficient estimates widely used in statistics.

The t-test for example is the maximum likelihood-based test to compare the mean of two normal distributions.

The MLE estimate for lm models leads to the **same solution** as the least squares (under given assumptions).

How to find the »best fit«

Solution:

Generally there is no closed solution (formula) for the parameters in function of the data

The point estimates are determined by a multi-step **iterative algorithms** (ex: Newton algorithm)

GLM diagnostic 1: Hat and Cook

- **Detection of influential observations and poor fitting**

- 1) **Hat values h_i** In analogy to LM there is a definition of a hat matrix for logistic regression fits and large diagonal values suggest a potential high influence (**leverage**) of a point on the obtained fit.
Limit $\sim 2p / n$ or $3p / n$.
- 2) **Cook's distance Cd_i** is a measure of a change in estimated coefficients when the observation i is ignored. Large values ($> \sim 4/n$) suggest a large influence, pointing to observations one might want to "investigate".

GLM diagnostic 2: Residuals

Response residuals:

$$r_i^{\text{resp}} = y_i - \hat{\mu}_i$$

- Raw difference between observed and fitted values.
- Not standardized.
- Not normally distributed in non-Gaussian GLMs.
- Useful for quick sense-check, but not for formal diagnostics.

Pearson residuals:

$$r_i^{\text{pearson}} = \frac{y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}$$

- Standardized by variance function.
 - Useful for spotting outliers. (Large |residual| (>2) suggests an outlier relative to model assumptions.)

GLM diagnostic 2: Residuals

Deviance residuals : $r_i^{\text{dev}} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{2 [\ell(y_i; y_i) - \ell(y_i; \hat{\mu}_i)]}$

- Based on likelihood.
- Measures how much worse the fitted value is compared to a “perfect” fit.
- Summed squares give overall deviance (see next)
- Preferred residual for GLMs.

```
Plots: influencePlot(model);  
residualPlot(model2, type = "response")  
residualPlot(model2, type = "pearson")  
residualPlot(model2, type = "deviance")
```

In general, None of these residuals are normally distributed!

GLM diagnostic 2: analysis of residuals

Dispersion check :Quantile Residuals QR_i

One can generate faked simulated data from the fitted model, like distribution of predicted new values and compare to the observed values (for each data point, each x_i).

The nb of fakes $< \text{obs}, y_i$ is called quantile residual Qr_i .

If the data are distributed as specified by the model these follow a uniform $U[0,1]$ distribution.

A Q-Q plot of calculated vs. expected quantile residuals can detect significant departures and suggest modifications to the model.

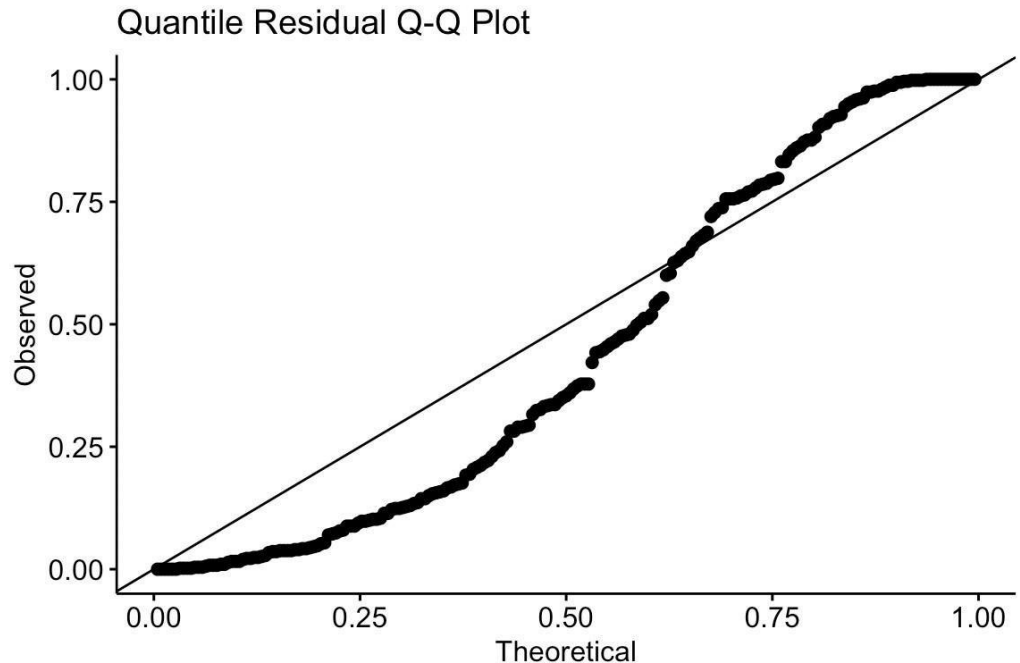
GLM diagnostic 2: analysis of residuals

Quantile Residuals QR_i

Example: (Poisson GLM QRs)

Several excessively extreme (larger and smaller than expected) QRs in this checking

suggests overdispersion of data (perhaps assumption on variance is not satisfied)



GLM diagnostic 6: autocorrelation

A quick test if there is any obvious evidence of **non-independence** of the observations:

Check for «autocorrelation», function `acf()`

Challenge 1

Using the babies dataset

- Fit a logistic regression to find parameters explaining the probability of prematurity ?
- What is the effect of birth weight on the probability of prematurity ?

Solution

```
> model2 <- glm(prem ~ bwt, family=binomial)
> summary(model2)
```

call:

```
glm(formula = prem ~ bwt, family = binomial)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.2879	-0.3985	-0.2784	-0.1810	3.0710

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.017338	0.717952	6.988	2.78e-12 ***
bwt	-0.067061	0.006808	-9.851	< 2e-16 ***

signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 664.66 on 1173 degrees of freedom
Residual deviance: 545.31 on 1172 degrees of freedom
AIC: 549.31

Number of Fisher scoring iterations: 6

Logistic regression: How to test significance and determine CI ?

Given standard errors SE of β :

test-statistics = estimate / SE = z

approx. Normal (under the null hypothesis)

called a **Wald-test**

CI width = approx. 1.96 * SE

CI symmetric for β and the log odds scale

⇒ not symmetric for the multiplicative effect $\exp(\beta)$ on the odds scale

⇒ not symmetric for the effect on the probability π

GLM diagnostics 3: Deviance

The deviance is the sum of the squared deviance residuals:

$$D = \sum_{i=1}^n (r_i^{\text{dev}})^2$$

Can also be seen as:

$$D(y; \hat{\theta}) = 2 \sum_{i=1}^n \left[\log L_i(\tilde{\theta}_i) - \log L_i(\hat{\theta}) \right].$$

Here:

- $L_i(\tilde{\theta}_i)$ is the likelihood contribution of observation i under the saturated model (fits y_i exactly),
- $L_i(\hat{\theta})$ is the likelihood contribution under the fitted model.

MLE likelihood and deviance

Highest (log) likelihood possible, predictors best adapted to each Y_i

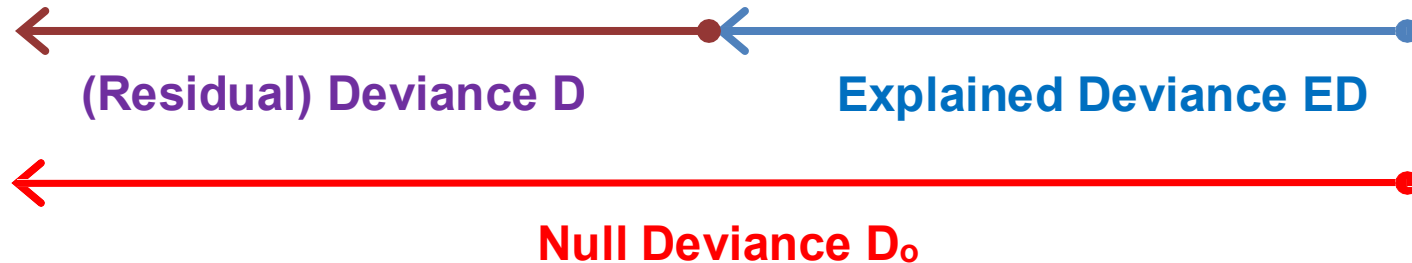
Deviance : difference in $2 \times \text{Log Lik}$

(log) likelihood if data from a fixed distribution with no individual observation-predictors

observed values,
Saturated model

fitted , current
proposed, Model

“Null model” ,
only 1 parameter



$$D_o = ED + D; \quad ED = D_o - D$$

$$R^2 \Rightarrow ED / D_o = (D_o - D) / D_o = 1 - (D / D_o) = \text{called a pseudo- } R^2$$

Notes: $\log \text{Lik} \leq 0$; good Log Lik is close to 0;

Deviance > 0 : a measure of "lack of fitting",

good is small positive close to 0

Parameter Optimization: Maximal (Log) Likelihood $\sim$ Minimal Deviance

GLM diagnostics 3: Deviance

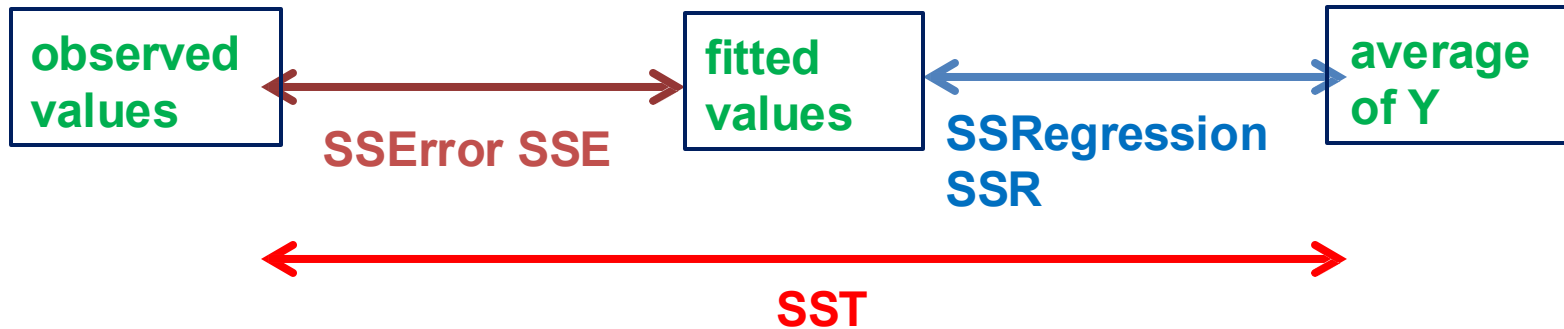
- ❑ **Null deviance:** how well (or bad) the response variable is predicted by a model that includes only the intercept (overall mean, logistic: binomial with fixed p) compared to the best possible model
- ❑ **Residual deviance:** how much deviance is missing compared to the best model after including the proposed set of independent variables (residual lack of fit)

Deviance: an analogy

$$SST = SSR + SSE$$

Total sum of squares = regression SS + residual SS

HEURISTIC REPRESENTATION



$$R^2 = SSR / SST = 1 - (SSE / SST)$$

Solution

```
> model2 <- glm(prem ~ bwt, family=binomial)
> summary(model2)
```

Call:

```
glm(formula = prem ~ bwt, family = binomial)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
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Multiple Logistic Regression

Extension to more than one predictor variable (either numeric or dummy variables).

With k predictors, the model is written:

$$\pi = \frac{e^{\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k}}{1 + e^{\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k}}$$

Adjusted Odds ratio for raising x_i by 1 unit, holding all other predictors constant:

$$OR_i = e^{\beta_i}$$

Solution

```
> model4 <- glm(prem ~ bwt*smoke+parity, family=binomial)
> summary(model4)
```

Call:

```
glm(formula = prem ~ bwt * smoke + parity, family = binomial)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.4798	-0.3998	-0.2784	-0.1682	2.9571

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	4.839354	0.978834	4.944	7.65e-07	***
bwt	-0.062082	0.008741	-7.103	1.22e-12	***
smokesmoker	2.247047	1.609071	1.396	0.1626	
paritynot first	-0.470085	0.283836	-1.656	0.0977	.
bwt:smokesmoker	-0.028043	0.015781	-1.777	0.0756	.

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(Dispersion parameter for binomial family taken to be 1)

Null deviance: 664.66 on 1173 degrees of freedom
Residual deviance: 532.93 on 1169 degrees of freedom
AIC: 542.93

Number of Fisher Scoring iterations: 6

model comparison tests



Likelihood Ratio Test LRT

Nested models

- the smaller model 1 is a special case of the larger model 2;
- larger model 2 has all predictors of model 1 and some additional predictors

Comparison of **nested** models:

- we can test if the improvement of model 2 over model 1 is statistically significant with a **likelihood ratio test (LRT) = deviance test = Wilks test**

model comparison tests

Likelihood Ratio Test LRT

Test statistic = $2 \times \text{Log Lik Ratio} = \text{Deviance D1} - \text{Deviance D2}$

~ **chi2 distribution** with degrees of freedom = df for smaller model (higher df)
- df for larger model

Example R code :

```
anova(model1, model2, test = "Chisq")
```

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
1		998		1200.5	
2	1	15.37	997	1185.1	8.9e-05 ***

The **same function** encodes the analogous model comparison test for LM models

Example for the class data :

```
anova(model.0, model.3)
```

Model 1: Height ~ Age

Model 2: Height ~ Age + Weight

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	17	1042.71				
2	16	524.94	1	517.77	15.781	0.0011 **

Model quality indices

R squared

Analogous of R^2 in LM

$$R^2 = \text{ED} / \text{D}_o = (\text{D}_o - \text{D}) / \text{D}_o = 1 - (\text{D} / \text{D}_o)$$

called a pseudo- R^2

Many different R-Squared and adjusted R-Squared have been proposed for GLM
Some are fairly widely used but generally model selection is best done with LRT

Akaike Information Criterion (AIC)

- allows to assess the quality of a model through comparison of related models
- based on the Deviance, but penalizes for the number of parameters (like adjusted R-squared, its intent is to correct for irrelevant predictors)

Solution

```
> model4 <- glm(prem ~ bwt*smoke+parity, family=binomial)
> summary(model4)
```

Call:

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Model selection

Nested Models: LRT !!!!

Otherwise: complicated
¿ Nothing simple works reliably ?

Multiple data methods (separate datasets learning-testing,
cross-validation, bootstraps)

(See: Day 4 & statistical learning / machine learning literature)